

ANALYTICAL PROBLEM SOLVING
CSA1021 CLOUD COMPUTING
& DATA ANALYTICS.Q1) Type-1 Hypervisor

Type-1 hypervisor is installed directly on the physical hardware without requiring a host operating system. It directly controls hardware resource and manage virtual machines.

Ex: VMWARE ESXi
Microsoft Hyper-V
XEN
KVM

Type-2 Hypervisor

Type 2 hypervisor runs on top of an existing operating system as windows or Linux. The host OS manages hardware resource and the hypervisor creates virtual machines within the operating system.

Ex: Oracle Virtual Box
Vmware workstation
Vmware Fusion.

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Directly on hardware	one top of Host OS
Performance	Very high	Moderate
Security	Excellent	Moderate
Resource utility	Efficient	less Efficient
Reliability	High	Moderate
Enterprise support	Excellent	Limited

Performance Analysis

⇒ Type-1 Hypervisors

→ communicate directly with physical hardware, reducing virtualization overhead.

→ CPU scheduling is fast, memory allocation is more efficient

→ Storage lower latency

→ Banking system, Healthcare, e-commerce

Type-2 Hypervisors

→ Rely on the host operating system for hardware access

→ The additional software layer increases CPU utilization and memory consumption leading to slow VM Execution.

Security Analysis

→ Type-1 hypervisors have a smaller attack surface because there is no host operating system. Features like secure boot, VM isolation, hardware assisted virtualization provide stronger

→ Type-2 hypervisor inherit vulnerabilities from the host operating system. If malware affects the host OS, all virtual machines running on it may also be compromised.

⇒ Mission critical applications require high Quality, availability, strong security, fast performance and scalability

Type-1 hypervisor satisfy these requirement because they provide minimal latency, efficient resource utilization, Strong VM isolation and Enterprise-grade management features

⇒ Type-1 hypervisor are the best choice because they offer superior performance, stronger security, better scalability, centralized management

Q.2)

Cloud Providers often host multiple customers on the same physical server to maximize resource utilization and reduce operational tasks.

The possible through Virtualization architecture

Virtualization Architecture for workload Isolation

Each VM has its own operating system, memory, CPU allocation, storage

CPU Isolation

- The hypervisor allocate dedicated CPU time
- One VM cannot directly access another VM's processor
- CPU scheduling ensure fair distribution

Memory Isolation

- Every VM its own Virtual memory.
- Memory Pages are protected by the hypervisor

Storage Isolation

- Each VM separate Virtual disk
- File system remain Independent

Device Isolation

- Virtual device such as NIC and storage controller are separately assigned.
- Drivers inside one VM cannot control hardware allocated to another VM.

Isolation of Different workload

- High security
- Encryption
- Strict Firewall policies
- Compliance with PCI-DSS

Major Attack Surface 2 : VM Escape Attack

Risk : unauthorized access.
Data leakage.

BA Protection

- updated hypervisor
- Least privilege access
- continuous security monitoring

Q3)

Study the following VM host utilization snapshot and recommend corrective actions

Introduction

The utilization value indicate that the VM host is operating close its maximum

High CPU usage, high memory utilization, high storage latency.

Analysis

CPU utilization: 92%.

Observation

- CPU usage is Extremely high.
- Very little processing capacity remains

Likely Bottlenecks.

- Excessive Virtual machine
- CPU contention
- Overload application

Corrective Actions

- Migrate VM using live migration
- Increase CPU cores
- Optimize application

Corrective Actions

- Increase RAM
- Remove unused VM
- configure memory reservation

Storage I/O latency 30ms

Observation

- Normal enterprise storage latency is below 10ms

Corrective Action

- upgrade SSD or NVME storage
- implement storage tiering

VM Boot time: 170 sec

VM startup is Very slow

Possible causes

- CPU congestion
- storage latency
- Heavy startup application

→ Perform live VM Migration

→ Add additional physical host

- upgrade processor and RAM.

Q4)

The Software Defined datacenter (SDDC) is a modern cloud architecture in which compute, storage, networking and security.

Traditional datacenter

- Physical servers.
- Manual provisioning
- Dedicated storage device

Software Defined Datacenter

- Virtual infrastructure
- Automated resource allocation
- Rapid provisioning
- High scalability

Compute Virtualization

- Multiple VM per server
- Live migration
- Rapid deployment
- Better hardware utilization

Storage Virtualization

- Centralized Management
- Dynamic storage allocation
- Storage migration

Benefits of SDDS

- Faster deployment
- Lower operational cost
- High flexibility
- Efficient resource utilization

Q.15)

The multi cloud federation connect multiple cloud providers to deliver same service

A common challenge is identity mismatch

Cause Identity Match

Different user ID format

Different authentication

Different authorization policies.

Problem caused

- Login failure
- Duplicate account
- unauthorized access
- Increase administrative overhead

Security identity design

- Federated identity Management

Benefits:

- Single Sign-on (SSO)
- centralized authentication

Single Sign on (SSO)

- improve usability
- reduced login failure
- Better productivity

Standard identity protocols

SAML

OAuth 2.0

OpenID connect (OIDC)

Multifactor Authentication

OTP

Authentication OTP

Biometric Verification